



Algebra Formulas Cheat Sheet

Complete algebra formulas cheat sheet with exponent rules, factoring patterns, quadratic formula, logarithm rules, and polynomial identities. Free PDF download.

Exponent Rules

NAME	FORMULA	NOTES
Product Rule	$a^m \cdot a^n = a^{m+n}$	Same base: add exponents
Quotient Rule	$\frac{a^m}{a^n} = a^{m-n}$	Same base: subtract exponents
Power Rule	$(a^m)^n = a^{mn}$	Power of a power: multiply
Product to Power	$(ab)^n = a^n b^n$	
Quotient to Power	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	
Zero Exponent	$a^0 = 1 \quad (a \neq 0)$	
Negative Exponent	$a^{-n} = \frac{1}{a^n}$	
Fractional Exponent	$a^{m/n} = \sqrt[n]{a^m}$	

Factoring Patterns

NAME	FORMULA	NOTES
Difference of Squares	$a^2 - b^2 = (a + b)(a - b)$	
Perfect Square (sum)	$a^2 + 2ab + b^2 = (a + b)^2$	
Perfect Square (diff)	$a^2 - 2ab + b^2 = (a - b)^2$	
Sum of Cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	
Difference of Cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	

Quadratic Equations

NAME	FORMULA	NOTES
Standard Form	$ax^2 + bx + c = 0$	
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
Discriminant	$D = b^2 - 4ac$	D > 0: two real roots D = 0: one root D < 0: complex roots
Vertex Form	$y = a(x - h)^2 + k$	Vertex at (h, k)
Sum of Roots	$r_1 + r_2 = -\frac{b}{a}$	
Product of Roots	$r_1 \cdot r_2 = \frac{c}{a}$	

Logarithm Rules

NAME	FORMULA	NOTES
Definition	$\log_a(x) = y \Leftrightarrow a^y = x$	
Product Rule	$\log_a(xy) = \log_a(x) + \log_a(y)$	
Quotient Rule	$\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$	
Power Rule	$\log_a(x^n) = n \log_a(x)$	
Change of Base	$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$	
Log of 1	$\log_a(1) = 0$	
Log of Base	$\log_a(a) = 1$	

Radical Rules

NAME	FORMULA	NOTES
Product	$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$	
Quotient	$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$	
Power	$\sqrt[n]{a^m} = a^{m/n}$	
Nested	$\sqrt[n]{\sqrt[m]{a}} = \sqrt[nm]{a}$	

